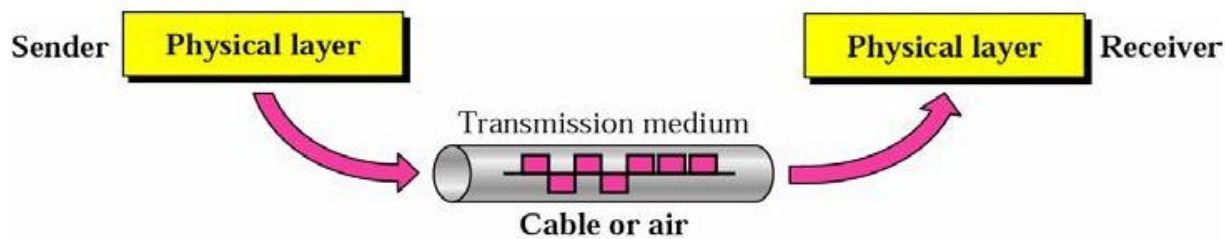


CHAPTER 2. TRANSMISSION MEDIA

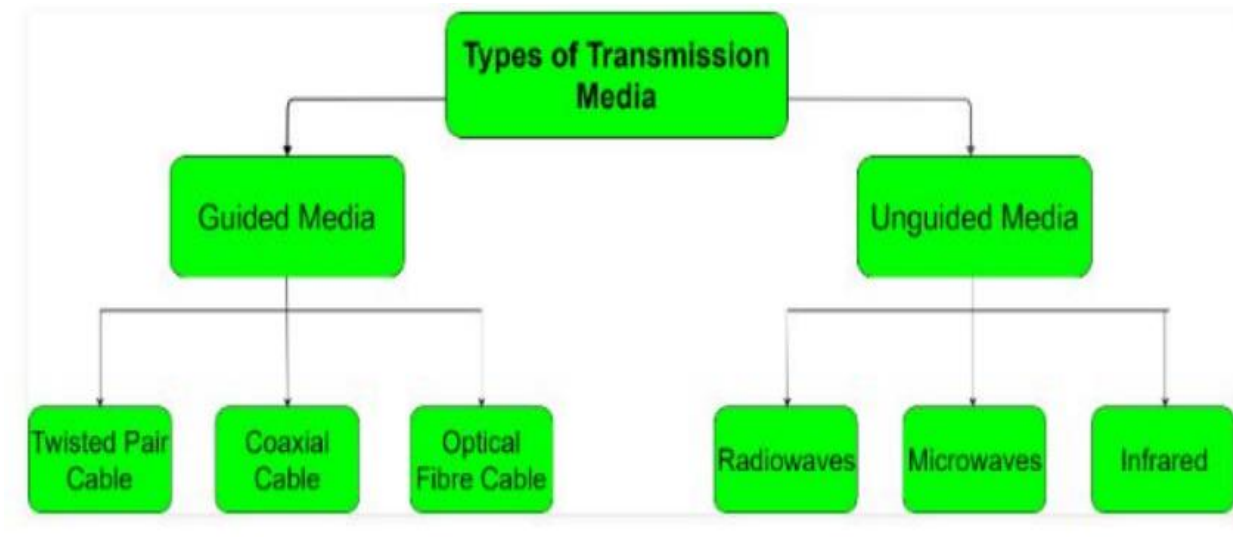
- Transmission media are located below the physical layer
- Computers use signals to represent data.
- Signals are retransmitted in form of electromagnetic energy.



WHAT IS TRANSMISSION MEDIA

- Transmission media is a pathway that carries the information from sender to receiver. We use different types of cables or waves to transmit data. Data is transmitted normally through electrical or electromagnetic signals.
- An electrical signal is in the form of current. An electromagnetic signal is series of electromagnetic energy pulses at various frequencies.
- These signals can be transmitted through copper wires, optical fibers, atmosphere, water and vacuum
- Different Medias have different properties like bandwidth, delay, cost and ease of installation and maintenance.

⊙ Transmission media is also called Communication channel.



Bound transmission media are the cables that are tangible or have physical existence and are limited by the physical geography.

⊙ Signals being transmitted are directed and confined in a narrow pathway by using physical links.

⊙ Popular bound transmission media in use are twisted pair cable, co-axial cable and fiber optical cable.

⊙ Each of them has its own characteristics like transmission speed, effect of noise, physical appearance, cost etc.

⊙ Features:

■ High Speed

■ Secure

■ Used for comparatively shorter distances .

 **WIRELESS OR UNGUIDED MEDIA OR UNBOUND TRANSMISSION MEDIA:**

☉ Unbound transmission media are the ways of transmitting data without using any cables.

☉ These media are not bounded by physical geography. This type of transmission is called Wireless communication.

☉ Nowadays wireless communication is becoming popular. Wireless LANs are being installed in office and college campuses.

☉ This transmission uses Microwave, Radio wave, Infrared are some of popular unbound transmission media.

☉ No physical medium is required for the transmission of electromagnetic signals.

☉ Features:

- Signal is broadcasted through air
- Less Secure
- Used for larger distances

FACTORS FOR TRANSMISSION MEDIAS

☉ The data transmission capabilities of various Medias vary differently depending upon the various factors. These factors are:

■ **1. Bandwidth**: It refers to the data carrying capacity of a channel or medium. Higher bandwidth communication channels support higher data rates.

■ **2. Radiation**: It refers to the leakage of signal from the medium due to undesirable electrical characteristics of the medium.

■ **3. Noise Absorption**: It refers to the susceptibility of the media to external electrical noise that can cause distortion of data signal.

■ **4. Attenuation:** It refers to loss of energy as signal propagates outwards. The amount of energy lost depends on frequency. Radiations and physical characteristics Of media contribute to attenuation.

BOUNDED OR GUIDED TRANSMISSION MEDIA

- ◎ Guided media, which are those that provide a conduit from one device to another, include Twisted-Pair Cable, Coaxial Cable, and Fibre-Optic Cable.
- ◎ A signal travelling along any of these media is directed and contained by the physical limits of the medium.
- ◎ Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current.
- ◎ Optical fibre is a cable that accepts and transports signals in the form of light.

TWISTED PAIR CABLE

- ◎ This cable is the most commonly used and is cheaper than others.
- ◎ It is lightweight, cheap, can be installed easily, and they support many different types of network.
- ◎ Some important points :
 - Its frequency range is 0 to 3.5 kHz.
 - Typical attenuation is 0.2 dB/Km @ 1kHz.
 - Typical delay is 50 μ s/km.
 - Repeater spacing is 2km.

- A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- One of these wires is used to carry signals to the receiver, and the other is used only as ground reference.
- The receiver uses the difference between the two.
- In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.

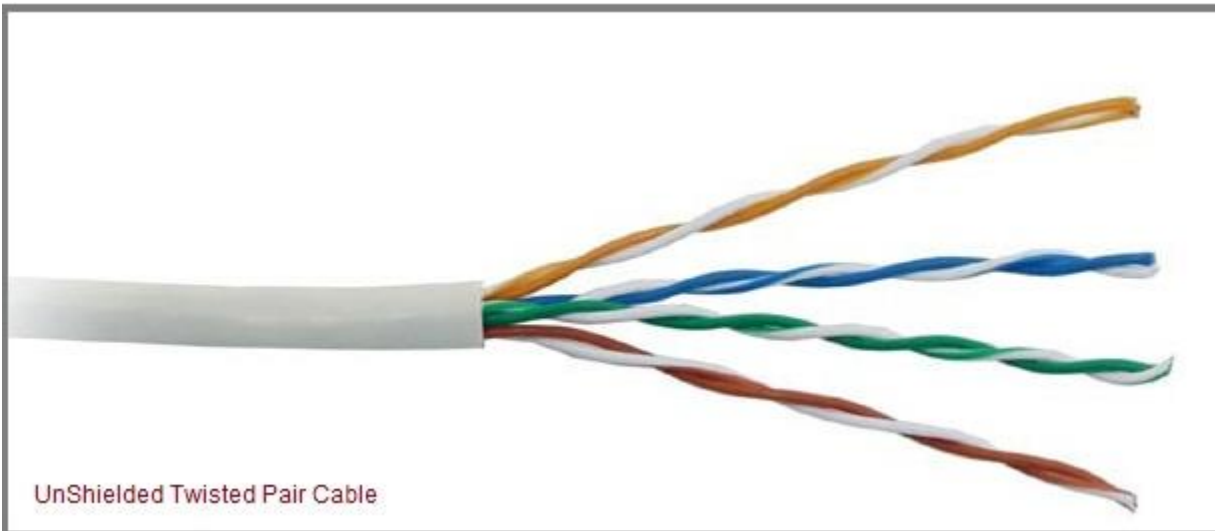
TWISTED PAIR IS OF TWO TYPES:

- ⦿ Unshielded Twisted Pair (UTP)
- ⦿ Shielded Twisted Pair (STP)

UNSHIELDED TWISTED PAIR CABLE

- ⦿ It is the most common type of telecommunication when compared with Shielded Twisted Pair Cable which consists of two conductors usually copper, each with its own colour plastic insulator.
- ⦿ Identification is the reason behind coloured plastic insulation.
- ⦿ UTP cables consist of 2 or 4 pairs of twisted cable. Cable with 2 pair use RJ-11 connector and 4 pair cable use RJ-45 connector.

UNSHIELDED TWISTED PAIR CABLE



ADVANTAGES OF UNSHIELDED TWISTED PAIR CABLE

- Installation is easy
- Flexible
- Cheap
- It has high speed capacity,
- 100 meter limit
- Higher grades of UTP are used in LAN technologies like Ethernet.
- It consists of two insulating copper wires (1mm thick). The wires are twisted together in a helical form to reduce electrical interference from similar pair.

DISADVANTAGES OF UNSHIELDED TWISTED PAIR CABLE

- Bandwidth is low when compared with Coaxial Cable

- ⦿ Provides less protection from interference.

SHIELDED TWISTED PAIR CABLE

- ⦿ This cable has a metal foil or braided-mesh covering which encases each pair of insulated conductors.
- ⦿ Electromagnetic noise penetration is prevented by metal casing.
- ⦿ Shielding also eliminates crosstalk
- ⦿ It has same attenuation as unshielded twisted pair.
- ⦿ It is faster than the unshielded and coaxial cable.
- ⦿ It is more expensive than coaxial and unshielded twisted pair



ADVANTAGES OF SHIELDED TWISTED PAIR CABLE

- ⦿ Easy to install
- ⦿ Performance is adequate
- ⦿ Can be used for Analog or Digital transmission
- ⦿ Increases the signaling rate
- ⦿ Higher capacity than unshielded twisted pair

- ⦿ Eliminates crosstalk

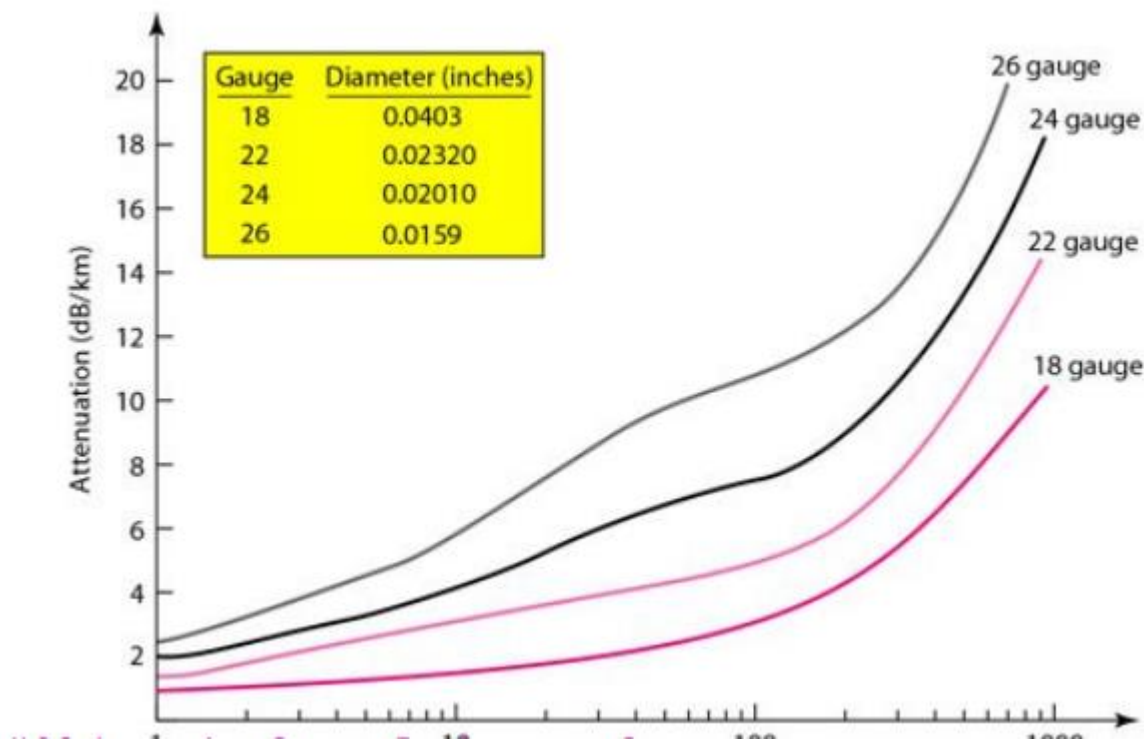
+DISADVANTAGES OF SHIELDED TWISTED PAIR CABLE

- ⦿ Difficult to manufacture

- ⦿ Heavy

+PERFORMANCE OF SHIELDED TWISTED PAIR CABLE

- ⦿ As shown in the below figure, a twisted-pair cable can pass a wide range of frequencies. However, with increasing frequency, the attenuation, measured in decibels per kilometre (dB/km), sharply increases with frequencies above 100kHz



+APPLICATIONS OF SHIELDED TWISTED PAIR CABLE

- ⦿ In telephone lines to provide voice and data channels.

- ⦿ The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.

⦿ Local Area Network, such as 10Base-T and 100Base-T, also use twisted-pair cables.

COAXIAL CABLE

⦿ Coaxial is called by this name because it contains two conductors that are parallel to each other.

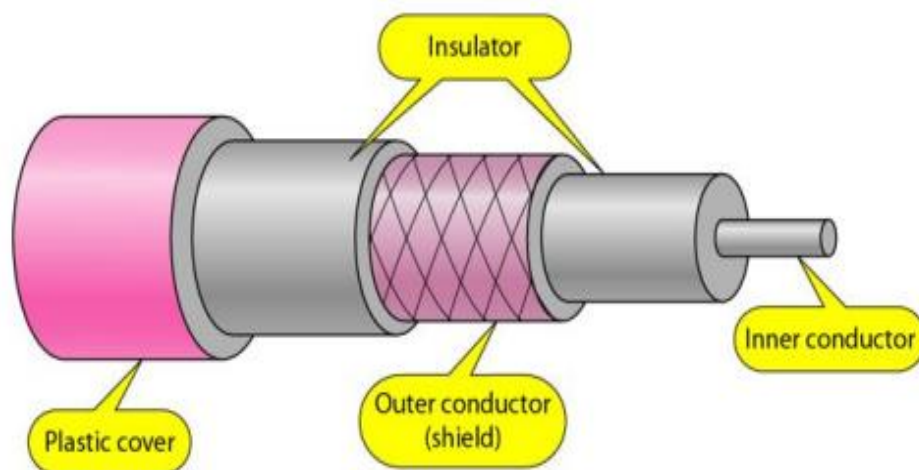
⦿ Copper is used in this as centre conductor which can be a solid wire or a standard one.

⦿ It is surrounded by PVC insulation, a sheath which is encased in an outer conductor of metal foil, braid or both.

⦿ Outer metallic wrapping is used as a shield against noise and as the second conductor which completes the circuit.

⦿ The outer conductor is also encased in an insulating sheath. The outermost part is the plastic cover which protects the whole cable

COAXIAL CABLE



HERE THE MOST COMMON COAXIAL STANDARDS

- ⦿ 50-Ohm RG-7 or RG-11 : used with thick Ethernet.
- ⦿ 50-Ohm RG-58 : used with thin Ethernet
- ⦿ 75-Ohm RG-59 : used with cable television
- ⦿ 93-Ohm RG-62 : used with ARCNET.

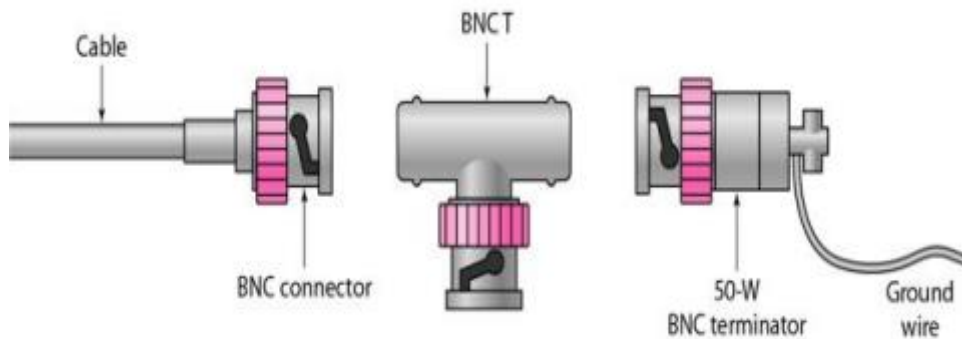
COAXIAL CABLE STANDARDS

- ⦿ Coaxial cables are categorized by their Radio Government(RG) ratings.
- ⦿ Each cable defined by an RG rating is adapted for a specialized function, as shown in the table below:

<i>Category</i>	<i>Impedance</i>	<i>Use</i>
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

COAXIAL CABLE CONNECTORS

- ⦿ To connect coaxial cable to devices, we need coaxial connectors.
- ⦿ The most common type of connector used today is the Bayonet Neill-Concelman (BNC) connector.
- ⦿ The below figure shows 3 popular types of these connectors: the BNC Connector, the BNC T connector and the BNC terminator.



- The BNC connector is used to connect the end of the cable to the device, such as a TV set.
- The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.
- The BNC terminator is used at the end of the cable to prevent the reflection of the signal.

THERE ARE TWO TYPES OF COAXIAL CABLES:

1. BaseBand

- This is a 50 ohm (Ω) coaxial cable which is used for digital transmission. It is mostly used for LAN's. Baseband transmits a single signal at a time with very high speed. The major drawback is that it needs amplification after every 1000 feet.

2. BroadBand

- This uses analog transmission on standard cable television cabling. It transmits several simultaneous signal using

different frequencies. It covers large area when compared with Baseband Coaxial Cable.

ADVANTAGES OF COAXIAL CABLE

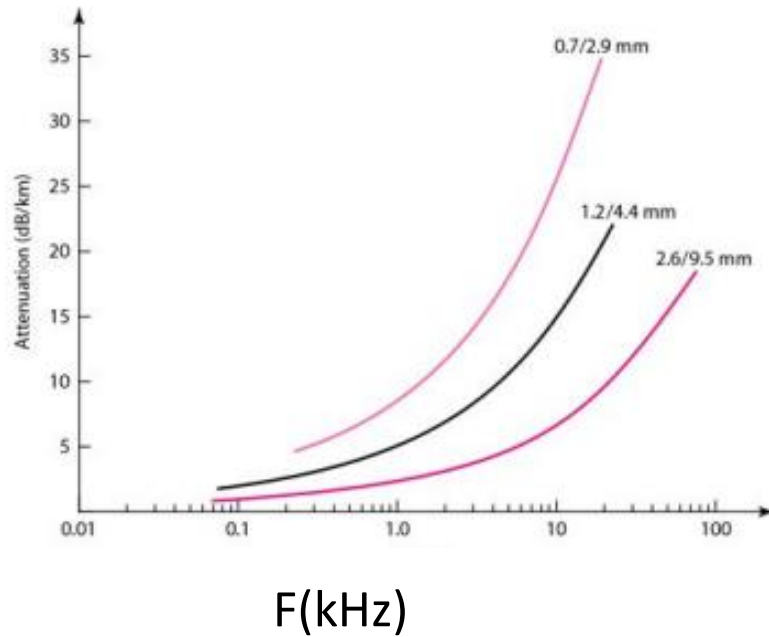
- ⦿ Bandwidth is high
- ⦿ Used in long distance telephone lines.
- ⦿ Transmits digital signals at a very high rate of 10Mbps.
- ⦿ Much higher noise immunity
- ⦿ Data transmission without distortion.
- ⦿ They can span to longer distance at higher speeds as they have better shielding when compared to twisted pair cable

DISADVANTAGES OF COAXIAL CABLE

- ⦿ Single cable failure can fail the entire network.
- ⦿ Difficult to install and expensive when compared with twisted pair.
- ⦿ If the shield is imperfect, it can lead to grounded loop.

PERFORMANCE OF COAXIAL CABLE

- ⦿ From the below figure, it can be seen that the attenuation is much higher in coaxial cable than in twisted-pair cable. In other words, although coaxial cable has a much higher bandwidth, the signal weakens rapidly and requires the frequent use of repeaters.



APPLICATIONS OF COAXIAL CABLE

- Coaxial cable was widely used in analog telephone networks, where a single coaxial network could carry 10,000 voice signals.
- Cable TV networks also use coaxial cables. In the traditional cable TV network, the entire network used coaxial cable. Cable TV uses RG-59 coaxial cable.
- In traditional Ethernet LANs. Because of its high bandwidth, and consequently high data rate, coaxial cable was chosen for digital transmission in early Ethernet LANs.
- The 10Base-2, or Thin Ethernet, uses RG-58 coaxial cable with BNC connectors to transmit data at 10Mbps with a range of 185 m.

FIBER OPTIC CABLE

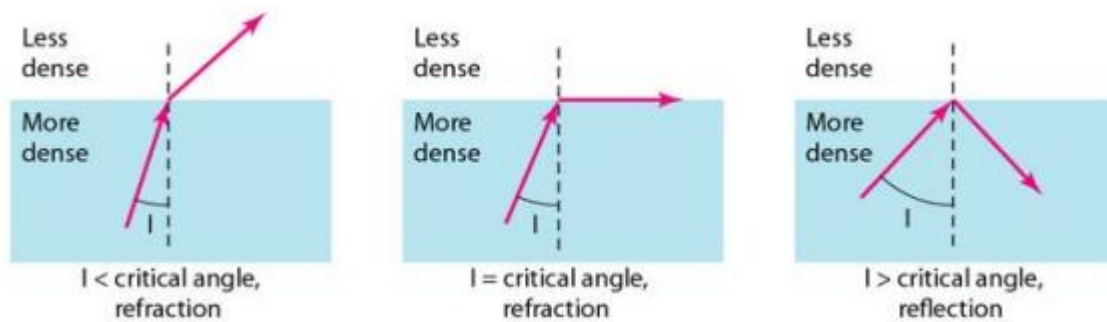
- A fibre-optic cable is made of glass or plastic and transmits signals in the form of light.

☉ For better understanding we first need to explore several aspects of the nature of light.

☉ Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light travelling through one substance suddenly enters another substance (of a different density), the ray changes direction.

☉ The below figure shows how a ray of light changes direction when going from a more dense to a less dense substance

BENDING OF A LIGHT RAY



AS THE FIGURE SHOWS:

☉ If the angle of incidence i (the angle the ray makes with the line perpendicular to the interface between the two substances) is less than the critical angle, the ray refracts and moves closer to the surface.

☉ If the angle of incidence is greater than the critical angle, the ray reflects (makes a turn) and travels again in the denser substance.

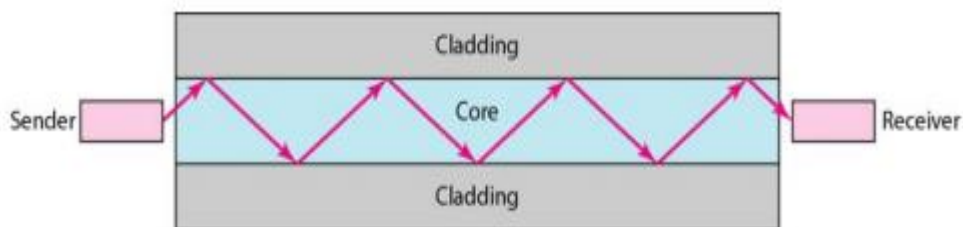
☉ If the angle of incidence is equal to the critical angle, the ray refracts and moves parallel to the surface as shown.

☉ Note: The critical angle is a property of the substance, and its value differs from one substance to another.

INTERNAL VIEW OF AN OPTICAL FIBRE

☉ Optical fibres use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic.

☉ The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.

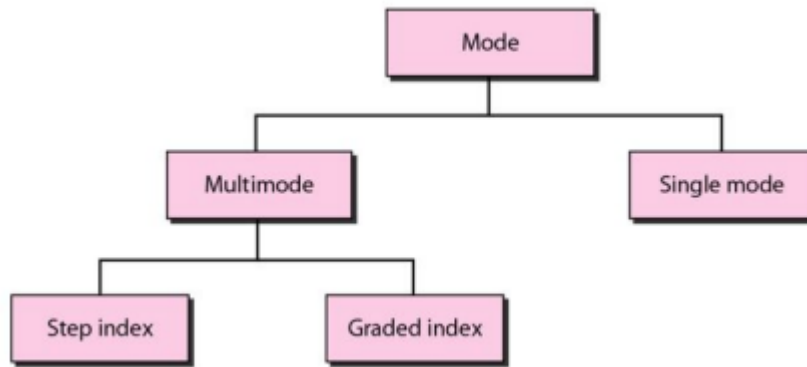


Internal view of an Optical fibre.

PROPAGATION MODES OF FIBER OPTIC CABLE

☉ Current technology supports two modes (Multimode and Single mode) for propagating light along optical channels, each requiring fibre with different physical characteristics.

☉ Multimode can be implemented in two forms: Step-index and Graded-index.

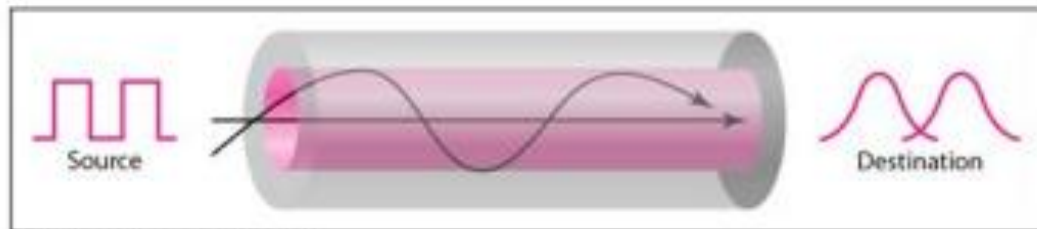


MULTIMODE PROPAGATION MODE

- Multimode is so named because multiple beams from a light source move through the core in different paths.
- How these beams move within the cable depends on the structure of the core as shown in the below figure.



a. Multimode, step index



b. Multimode, graded index



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MULTIMODE PROPAGATION MODE

- ⦿ In multimode step-index fibre, the density of the core remains constant from the centre to the edges.
- ⦿ A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.
- ⦿ The term step-index refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fibre.
- ⦿ In multimode graded-index fibre, this distortion gets decreases through the cable.
- ⦿ The word index here refers to the index of refraction. This index of refraction is related to the density.
- ⦿ A graded-index fibre, therefore, is one with varying densities. Density is highest at the centre of the core and decreases gradually to its lowest at the edge.

SINGLE MODE

- ⦿ Single mode uses step-index fibre and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.
- ⦿ The single-mode fibre itself is manufactured with a much smaller diameter than that of multimode fibre, and with substantially lower density.

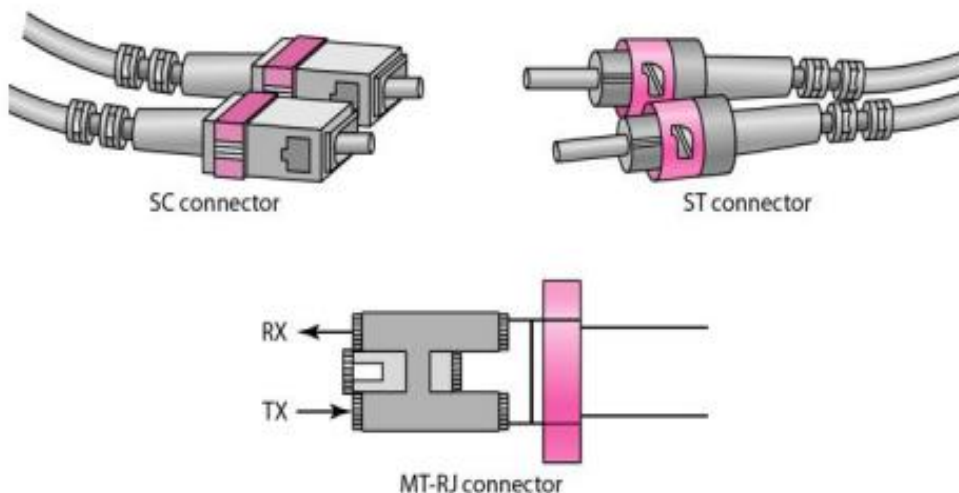
⦿ The decrease in density results in a critical angle that is close enough to 90 degree to make the propagation of beams almost horizontal.

✚ FIBRE SIZES FOR FIBER OPTIC CABLE

⦿ Optical fibres are defined by the ratio of the diameter or their core to the diameter of their cladding, both expressed in micrometers. The common sizes are shown in the figure below:

Type	Core (μm)	Cladding (μm)	Mode
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

✚ FIBRE OPTIC CABLE CONNECTOR



The Subscriber Channel(SC) connector is used for cable TV. It uses push/pull locking system. The Straight-Tip(ST) connector is used for connecting cable to the networking devices. MT-RJ is a connector that is the same size as RJ45.

ADVANTAGES OF FIBRE OPTIC CABLE

☉ Fibre optic has several advantages over metallic cable:

- Higher bandwidth
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials
- Light weight
- Greater immunity to tapping.

DISADVANTAGES OF FIBRE OPTIC CABLE

☉ There are some disadvantages in the use of optical fibre:

- Installation and maintenance
- Unidirectional light propagation
- High Cost

PERFORMANCE OF FIBRE OPTIC CABLE

☉ Attenuation is flatter than in the case of twisted-pair cable and coaxial cable.

☉ The performance is such that we need fewer(actually one tenth as many) repeaters when we use the fibre-optic cable

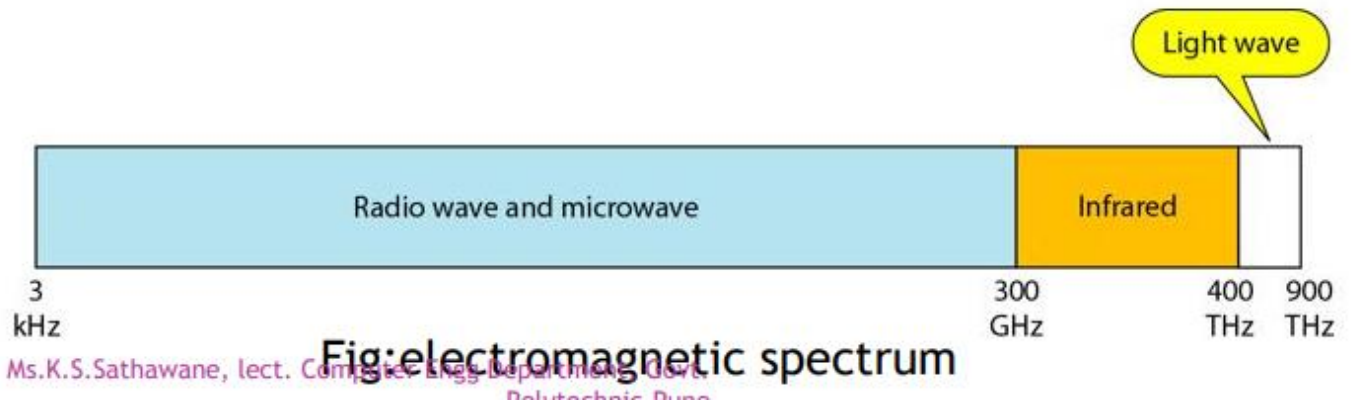
APPLICATIONS OF FIBRE OPTIC CABLE

- ⦿ Often found in backbone networks because its wide bandwidth is cost-effective.
- ⦿ Some cable TV companies use a combination of optical fibre and coaxial cable thus creating a hybrid network.
- ⦿ Local-area Networks such as 100Base-FX network and 1000Base-X also use fibre-optic cable.

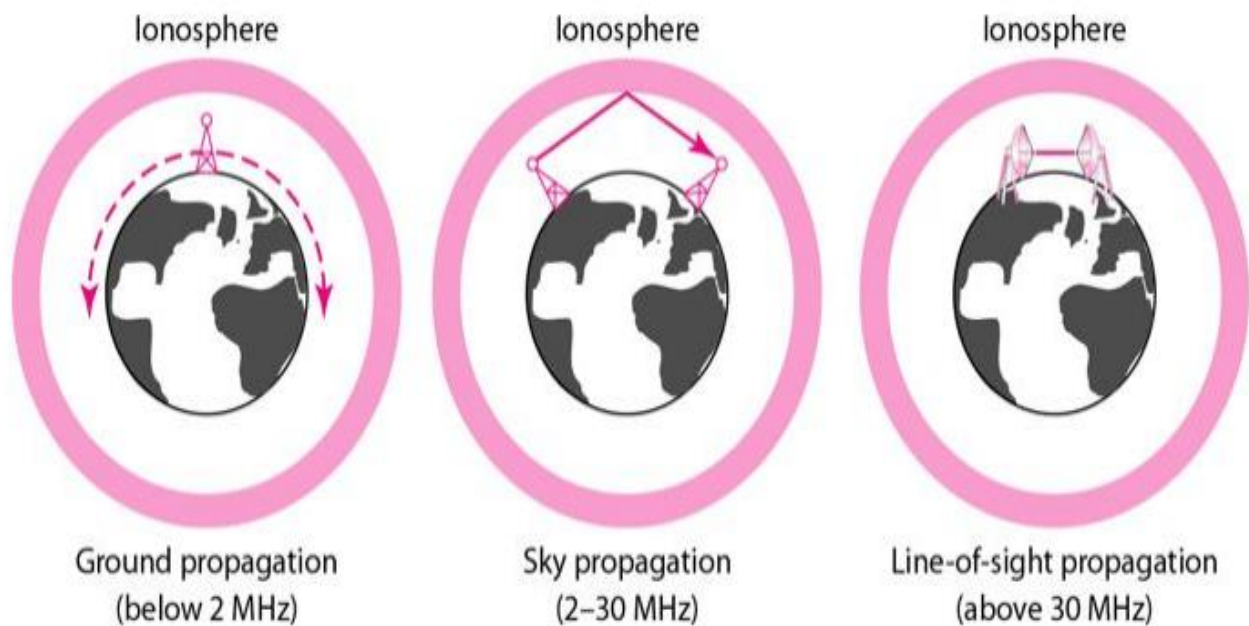
➤ UNBOUNDED OR UNGUIDED TRANSMISSION MEDIA

UNBOUNDED OR UNGUIDED TRANSMISSION MEDIA

- ⦿ Unguided medium transport electromagnetic waves without using a physical conductor.
- ⦿ This type of communication is often referred to as wireless communication.
- ⦿ Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.
- ⦿ The below figure shows the part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz, used for wireless communication.



☉ Unguided signals can travel from the source to the destination in several ways: Ground propagation, Sky propagation and Line-of-sight propagation as shown in below figure :



PROPAGATION MODES

☉ Ground Propagation: In this, radio waves travel through the lowest portion of the atmosphere, hugging the Earth.

- ☉ These low-frequency signals radiate in all directions from the transmitting antenna and follow the curvature of the planet.
- ☉ Sky Propagation: In this, higher-frequency radio waves radiate upward into the ionosphere where they are reflected back to Earth.
- ☉ This type of transmission allows for greater distances with lower output power.
- ☉ Line-of-sight Propagation: in this type, very high-frequency signals are transmitted in straight lines directly from antenna to antenna.

WE CAN DIVIDE WIRELESS TRANSMISSION INTO THREE BROAD GROUPS:

- ☉ Radio waves
- ☉ Micro waves
- ☉ Infrared waves



RADIO WAVES

- ☉ Electromagnetic waves ranging in frequencies between 3 KHz and 1 GHz are normally called radio waves.
- ☉ Radio waves are omnidirectional. When an antenna transmits radio waves, they are propagated in all

directions. This means that the sending and receiving antennas do not have to be aligned.

⦿ A sending antenna send waves that can be received by any receiving antenna. The omnidirectional property has disadvantage, too.

⦿ The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal using the same frequency or band.

⦿ Radio waves, particularly with those of low and medium frequencies, can penetrate walls. This characteristic can be both an advantage and a disadvantage.

⦿ It is an advantage because, an AM radio can receive signals inside a building. It is a disadvantage because we cannot isolate a communication to just inside or outside a building.



OMNIDIRECTIONAL ANTENNA FOR RADIO WAVES





APPLICATIONS OF RADIO WAVES

- ◎ The omnidirectional characteristics of radio waves make them useful for multicasting in which there is one sender but many receivers.
- ◎ AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.



MICRO WAVES

- ◎ Electromagnetic waves having frequencies between 1 and 300 GHz are called micro waves. Micro waves are unidirectional.
- ◎ When an antenna transmits microwaves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.
- ◎ The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.

THE FOLLOWING DESCRIBES SOME CHARACTERISTICS OF MICROWAVES PROPAGATION:

- ◎ Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall.

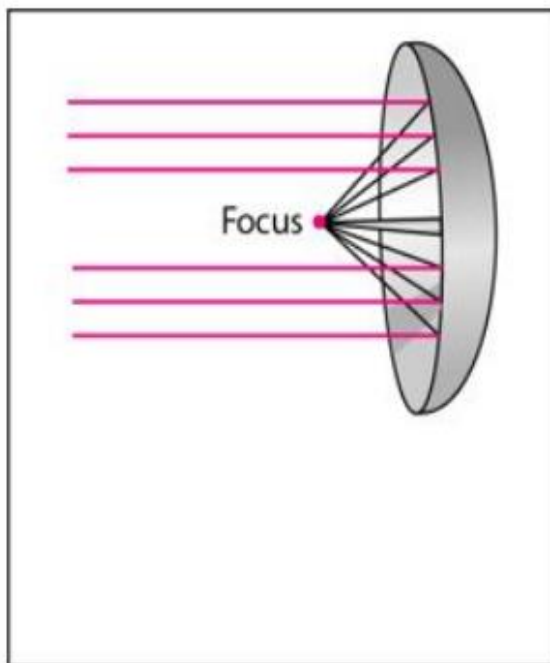
⦿ Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside the buildings.

⦿ The microwave band is relatively wide, almost 299 GHz. Therefore, wider sub-bands can be assigned and a high data rate is possible.

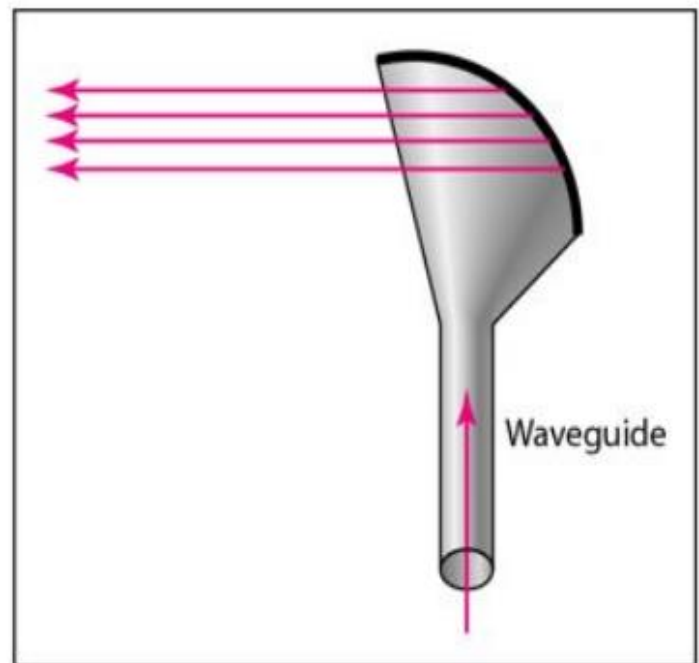
⦿ Use of certain portions of the band requires permission from authorities



UNIDIRECTIONAL ANTENNA FOR MICRO WAVES



a. Dish antenna



b. Horn antenna

⦿ A parabolic antenna works as a funnel, catching a wide range of waves and directing them to a common point.

⦿ In this way, more of the signal is recovered than would be possible with a single-point receiver.

⦿ A horn antenna looks like a gigantic scoop. Outgoing transmissions are broadcast up a stem and deflected outward in a series of narrow parallel beams by the curved head.

⦿ Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem



APPLICATIONS OF MICRO WAVES

⦿ Microwaves, due to their unidirectional properties, are very useful when unicast(one-to-one) communication is needed between the sender and the receiver. They are used in cellular phones, satellite networks and wireless LANs.

⦿ There are 2 types of Microwave Transmission

- Terrestrial Microwave
- Satellite Microwave
-

▪ ADVANTAGES OF MICROWAVE TRANSMISSION

⦿ Used for long distance telephone communication

⦿ Carries 1000's of voice channels at the same time



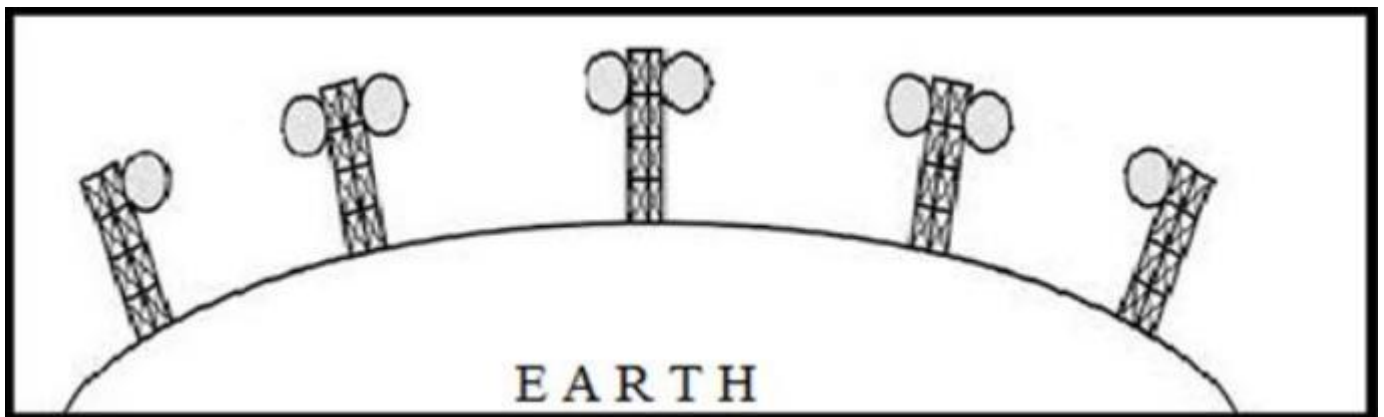
DISADVANTAGES OF MICROWAVE TRANSMISSION

⦿ It is very costly



TERRESTRIAL MICROWAVE

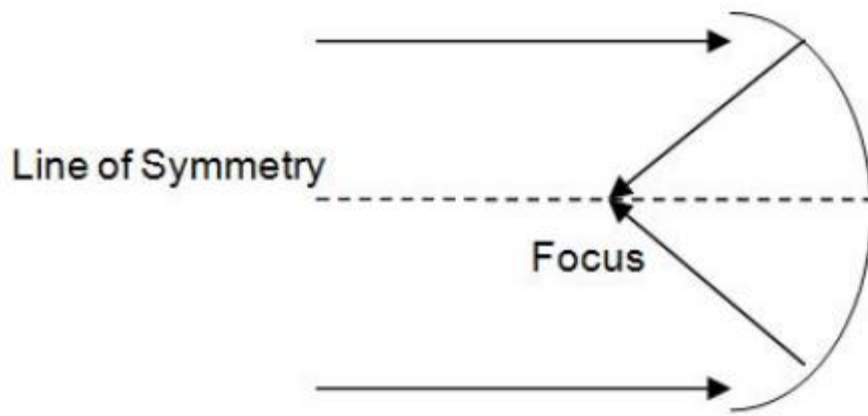
- ⦿ For increasing the distance served by terrestrial microwave, repeaters can be installed with each antenna .
- ⦿ The signal received by an antenna can be converted into transmittable form and relayed to next antenna as shown in below figure.
- ⦿ It is an example of telephone systems all over the world



There are two types of antennas used for terrestrial microwave communication :

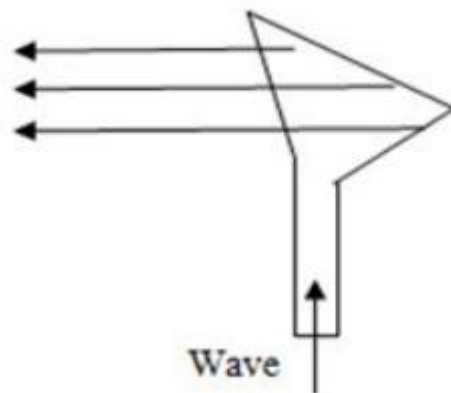
1. PARABOLIC DISH ANTENNA

- ⦿ In this every line parallel to the line of symmetry reflects off the curve at angles in a way that they intersect at a common point called focus.
- ⦿ This antenna is based on geometry of parabola.



2. HORN ANTENNA

- It is a like gigantic scoop.
- The outgoing transmissions are broadcast up a stem and deflected outward in a series of narrow parallel beams by curved head.



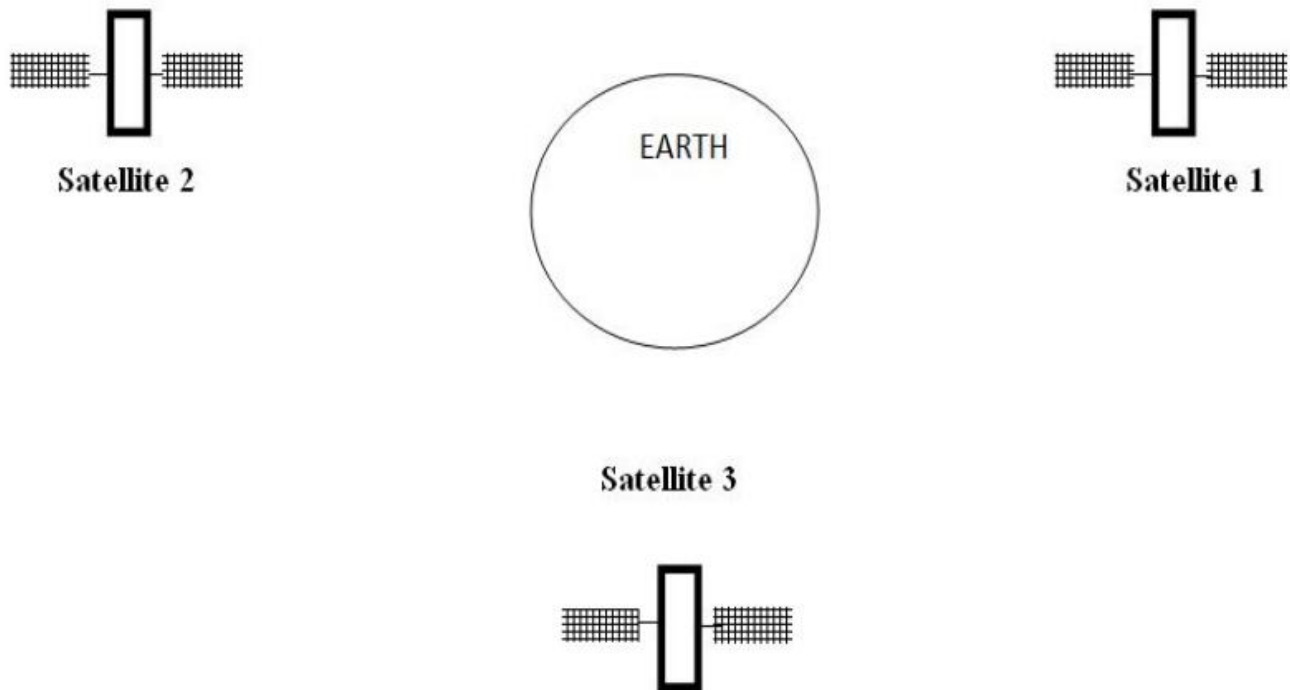
SATELLITE MICROWAVE

- This is a microwave relay station which is placed in outer space.
- The satellites are launched either by rockets or space shuttles carry them.

☉ These are positioned 36000 Km above the equator with an orbit speed that exactly matches the rotation speed of the earth.

☉ As the satellite is positioned in a geo-synchronous orbit, it is stationery relative to earth and always stays over the same point on the ground.

☉ This is usually done to allow ground stations to aim antenna at a fixed point in the sky.



FEATURES OF SATELLITE MICROWAVE

☉ Bandwidth capacity depends on the frequency used.

☉ Satellite microwave deployment for orbiting satellite is difficult.



ADVANTAGES OF SATELLITE MICROWAVE

- ⦿ Transmitting station can receive back its own transmission and check whether the satellite has transmitted information correctly.
- ⦿ A single microwave relay station which is visible from any point.



DISADVANTAGES OF SATELLITE MICROWAVE

- ⦿ Satellite manufacturing cost is very high
- ⦿ Cost of launching satellite is very expensive
- ⦿ Transmission highly depends on whether conditions, it can go down in bad



INFRARED WAVES

- ⦿ Infrared waves, with frequencies from 300 GHz to 400 THz, can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls.
- ⦿ This advantageous characteristic prevents interference between one system and another, a short-range communication system in one room cannot be affected by another system in the next room.
- ⦿ When we use infrared remote control, we do not interfere with the use of the remote by our neighbours.

⦿ However, this same characteristic makes infrared signals useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication

APPLICATIONS OF INFRARED WAVES

⦿ The infrared band, almost 400 THz, has an excellent potential for data transmission.

⦿ Such a wide bandwidth can be used to transmit digital data with a very high data rate.

⦿ The Infrared Data Association(IrDA), an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mouse, PCs and printers.

⦿ Infrared signals can be used for short-range communication in a closed area using line-of-sight